

Sai Karthik Battula

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Education

University of Texas at Dallas

May 2027

B.S. Computer Science ~ GPA 3.96 | Dean's List

Projects

Samsa – AI-Augmented Prediction Market Platform

Dec 2025

- Migrated data storage from flat JSON to PostgreSQL, improving query performance and enabling structured analysis of market data at scale.
- Refactored the React frontend and enhanced Node.js/Express backend, improving component architecture, state management, and API routing.

AI Agent Marketplace

Jun 2025

- Co-founded a Fiverr-style marketplace platform for AI agents, enabling users to discover, deploy, and chain agents for real-world task automation.
- Designed and implemented backend architecture including MCP server setup and core agent orchestration logic for sequential task execution.
- Built tools enabling agents to automatically retrieve structured data from LinkedIn and external sources to execute multi-step real-world workflows.

Experience

Nebula Labs — Notebook Platform

Feb 2026

- Contributing to a collaborative notebook MVP featuring login, class search, note uploads, and profile management for student knowledge-sharing.
- Optimized database lookups for course and professor notes, reducing query latency by 40% and enabling faster retrieval by teacher or class name.

Engineering Projects in Community Service – UT Design

Aug – Dec 2025

- Curated and labeled 1,900+ indoor images to build a high-quality training dataset for a YOLO object detection model targeting indoor landmarks and obstacle recognition.
- Trained, evaluated, and iteratively optimized the YOLO model, achieving robust detection performance on a diverse indoor environment dataset.
- Quantized and converted the model to TensorFlow Lite (TFLite), enabling real-time, on-device inference with reduced memory footprint — critical for deployment on embedded navigation hardware.
- Integrated the ML model with the navigation engine, directly improving indoor positioning accuracy and enabling reliable landmark-based navigation for users.
- Won a Design of Excellence Award for coordinating cross-timezone collaboration with an international engineering team, helping turn an emerging project with an uncertain direction into a meaningful, progressing solution.

Software Engineering Fellow – Headstarter AI

Jun – Aug 2024

- Built an AI-driven customer support agent using OpenAI and Pinecone vector search, improving query response accuracy by 40% over baseline keyword matching.
- Engineered a SaaS flashcard product powered by Llama 3.1 for dynamic content generation; integrated Stripe for subscription-based monetization.
- Designed a web scraper feeding live data into a LangChain RAG pipeline to provide real-time, context-aware responses grounded in up-to-date information.

Skills

AI/ML: PyTorch/TensorFlow, TFLite, YOLO, LangChain, OpenAI API, Pinecone, Scikit-learn, NumPy, RAG pipelines, model quantization

Languages: Python, JavaScript, C++, Java, SQL, HTML/CSS

Frameworks & Tools: React.js, Next.js, Node.js, Express, Flask, Firebase, PostgreSQL, Git/GitHub, CI/CD, Jupyter, Matplotlib

Certifications: Machine Learning Specialization, CCNA: Introduction to Networks, IT Specialist (Networking, Cybersecurity, Network Security)